INFINITIX

BEST AI Ready Enterprise
---AI-Stack

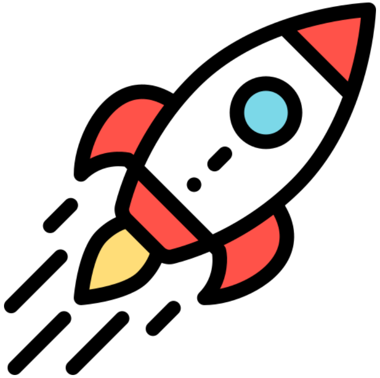
The next leading platform for all AI development、Inference and operations.

Allen Chen

Agenda

1. Infinitix Profile
2. Solution Proposal
3. AI-Stack Feature
4. AI-Stack Live Demo

Infinitix Profile



Company's Growth Journey - Review & Outlook

Before 2019

- Launched AI-Stack

NCHC NARLabs 財團法人國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing

2019

NVIDIA INCEPTION PROGRAM

2021

NVIDIA PREFERRED SOLUTION ADVISOR

2024

- **AICC -** 1st Computing Center in Asia with H200 *128

2025

moda 數位發展部 Ministry of Digital Affairs

- HGX *4 + AMD MI300X *1

Many leading companies develop and operate AI with AI-Stack.

Semi-conductor

UMC
Himax
fitipower
SiS
REALTEK

Energy

台湾電力公司

Transportation

Taoyuan Airport
桃園機場
Metro Taipei
metro Taoyuan

Manufacturing

中鋼公司
台塑企業
FORMOSA PLASTICS GROUP
DELTA

Government

NARLabs 國家實驗研究院
國家高速網路與計算中心
National Center for High-performance Computing
CENTRAL WEATHER BUREAU
金門縣政府

Finance Service

永豐金控
SinoPac Holdings
iPASS 一卡通

Healthcare

Academic

中央研究院
ACADEMIA SINICA
TAIPEI TECH

Over Sea

UNION TOOL
TEL
FLEXINFRA
COREX
HITACHI
Inspire the Next

Teleport

Ecosystem

Channel

MACNICA, KIWI technology Inc., Toontec Solutions

Channel

ZERONE 零壹科技, Stark Technology Inc. 敦陽科技股份有限公司, Enlight 翔耀實業股份有限公司, UNIFOSA, MACNICA, Toontec Solutions, 敦新科技 DAWNING TECHNOLOGY INC., 豐康科技 www.fongcon.com.tw, APEX FONG YI TECHNOLOGY CO., LTD. 峰儀科技股份有限公司, tsti 大世科, 無敵 BESTA, 科智企業 YOUR UPTIME IS OUR FIRST PRIORITY

Channel

FLEXINFRA

INFINITIX

Server Partner

NVIDIA, DELL EMC, Hewlett Packard Enterprise, AMD, SUPERMICRO, GIGABYTE

Storage Partner

NetApp, DELL EMC, NUTANIX, MINIO, BIGTERA, accelstor

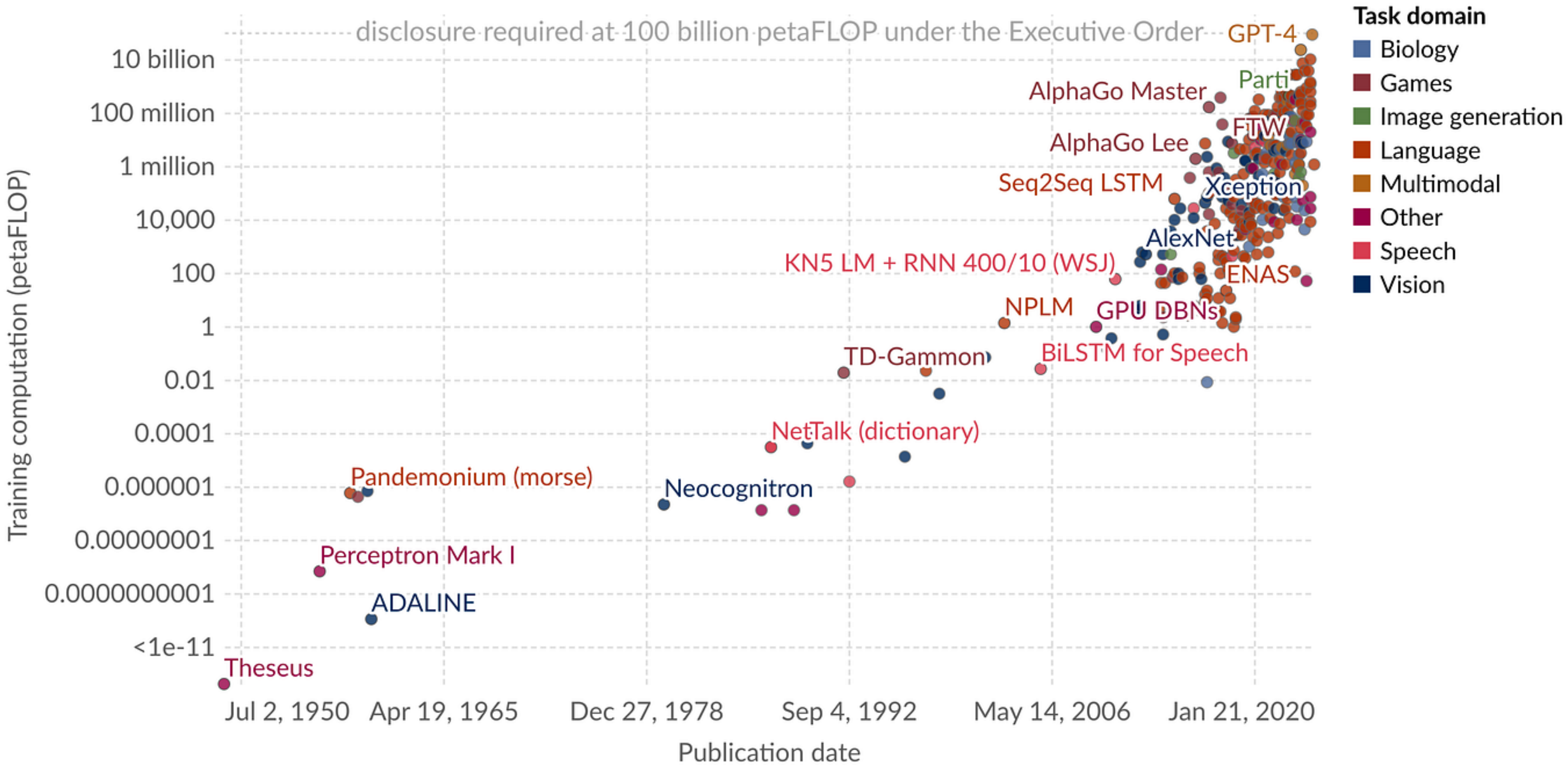
Edge Partner

ADLINK LEADING EDGE COMPUTING, MACNICA, aetina

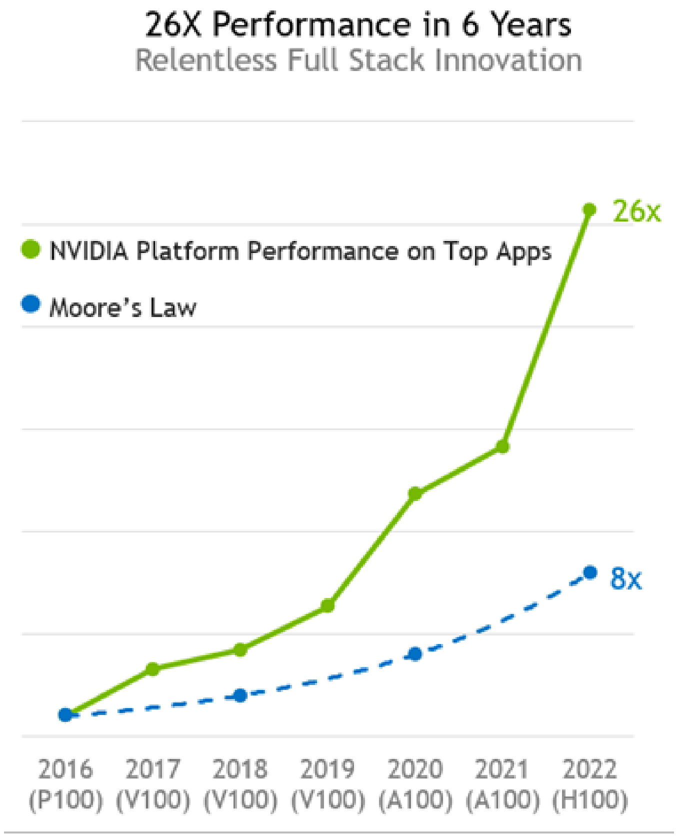
AI Application Partner

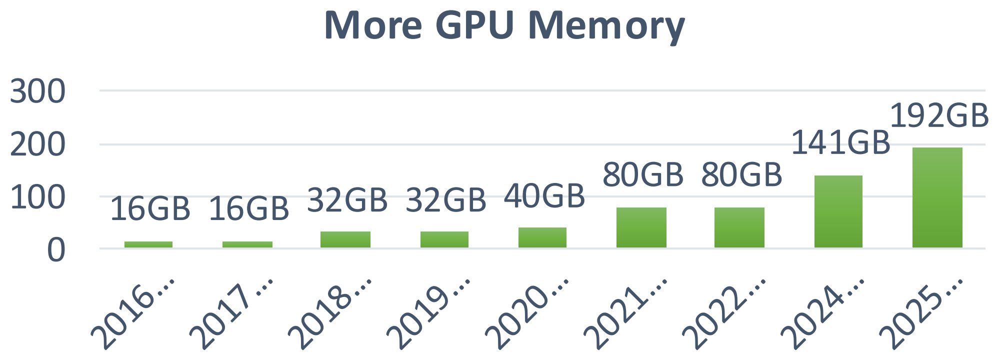
GRAID, POWER ARENA, PROFET AI Realize AI within one week!, PEGATRON 和碩聯合科技, is-land, Sightour, MUEN, 科智企業, LEDA TECHNOLOGY

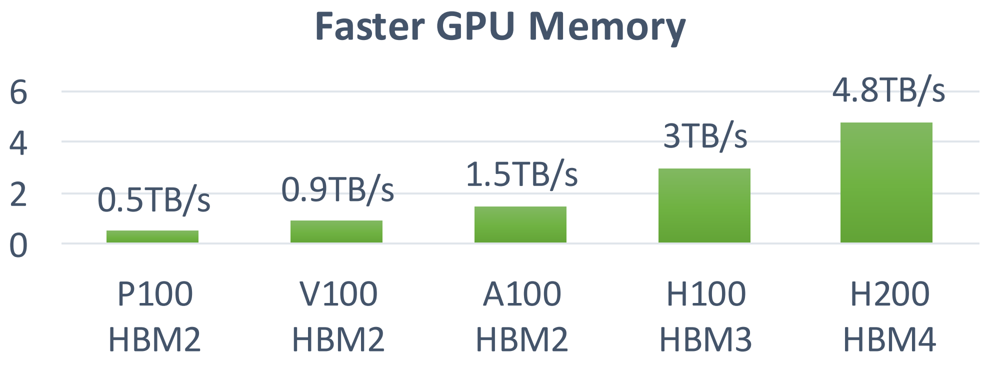
Rapid Growth in AI Computing Demand



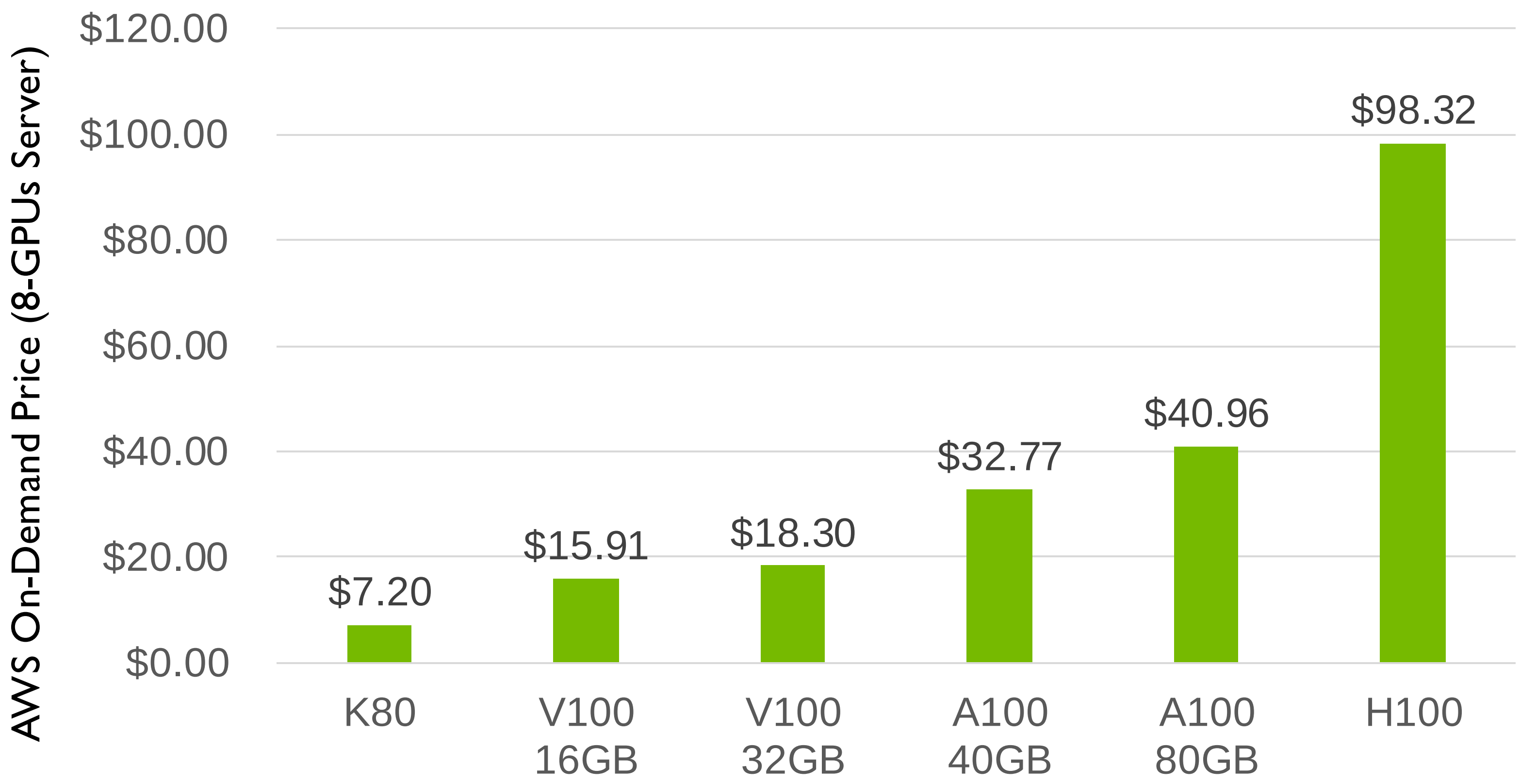
The World Enters an Era of High-Performance AI Infrastructure



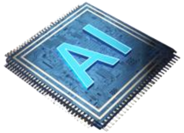




GPU Procurement Costs Increased 10x



Evolution of artificial intelligence applications



Step One

Training
Fine Tune

Step Two

Inference

Identify
Predict

Step Three

GenAI

Generative AI is a branch of artificial intelligence, mainly used for creative work, such as article generation, image generation, music generation, etc.

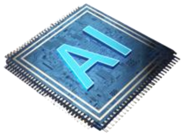
Step Four

Agent AI

AI Agent refers to artificial intelligence that can make decisions and complete actions independently without human intervention.

GPU computing power situation

✓ A lot GPU Resource
✓ GPU Aggregation
✓ Elastic Training



Step One

Training

Fine Tune

Step Two

推論服務

辨識、預測

GenAI

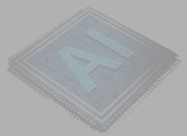
生成式AI是人工智慧中的一個分支，主要用於創造性的工作，例如文章生成、影像生成、音樂生成等。

Four

nt AI

指的是能夠自 決策、完成行動，而且毋需人類介入的人工智慧。

GPU computing power situation



Step One

模型訓練

Step Two

Inference

Identify
Predict

- ✓ <u>Small</u> GPU resource
- ✓ GPU Partitioning

GenAI

生成式AI是人工智慧中的一個分支，主要用於創造性的工作，例如文章生成、影像生成、音樂生成等。

自
主做出決策、完成行動，而且毋需人類介入的人工智慧。

GPU computing power situation

- ✓ Normalization GPU Resource
- ✓ GPU Partitioning

Step Two

推論服務

辨識、預測

Step One

模型訓練

Step Three

GenAI

Generative AI is a branch of artificial intelligence, mainly used for creative work, such as article generation, image generation, music generation, etc.

Step Four

Agent AI

AI Agent，指的是能夠自主做出決策、完成行動，而且毋需人類介入的人工智慧。

GPU computing power situation

- ✓ GPU hybrid allocation
- ✓ GPU Partitioning +Aggregation
- ✓ GPU scheduling strategy

Step One

模型訓練

Step Two

推論服務

辨識、預測

生成式AI是人工智慧中的一個分支，主要用於創造性的工作，例如文章生成、影像生成、音樂生成等。

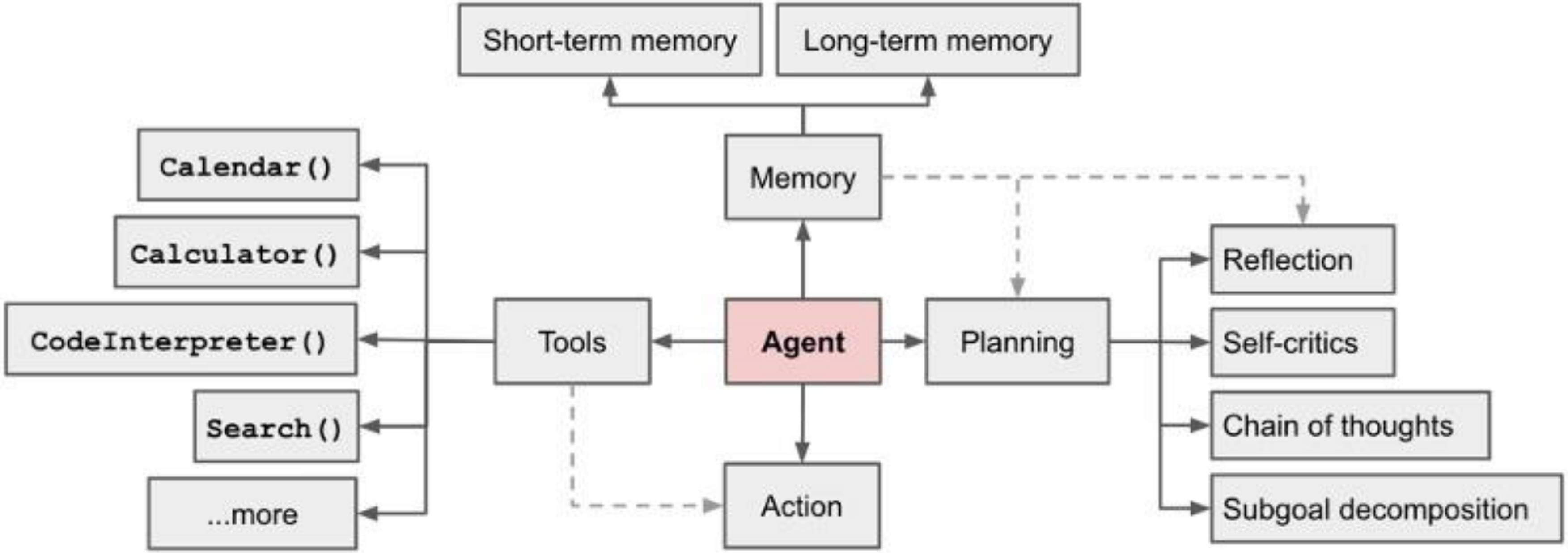
Step Four

Agent AI

AI Agent refers to artificial intelligence that can make decisions and complete actions independently without human intervention.

After AI ---AI Agent

AI Agent uses LLM as its brain, has planning skills, has memory, and can call tools (tool use). Therefore, AI Agent has a way to solve general problems, which is no longer comparable to previous dedicated artificial intelligence.



Taken from the blog of Lillian Weng/Head of security systems at OpenAI and a research scientist who has led AI application research

Ian Buck, Vice President of Nvidia, said that major server service providers are developing rapidly in three major areas of AI:

1. Infrastructure: Provide GPU computing power to AI startups and developer communities at a very large scale. Cloud vendors get high returns from their customers' GPU investments.

2. Token serving (Language Model as a Service): Allow customers to use large language models such as GPUs and LLaMA. For every $1 invested, you can get $7 over four years.

3. Building a new generation of ultra-large-scale AI models: Extensive infrastructure and algorithm innovation are required. In the future, the scale of training will be from hundreds of thousands of GPUs to millions of GPUs.

AI and GPU Computing Power

Training **Inference**

GenAI **Agent AI**

✖

GPU

Different GPU for different Workload

	GPU	Memory (VRAM)	DL Training & DA	DL Inference	HPC / AI	Omniverse/ Render Farms	Virtual Workstation	Virtual Desktop(VDI)	Edge Accleration
Compute	HGX H100	80GB HBM3 per GPU	SXM ★★★	SXM ★★★	SXM ★★★				
	H100 NVL	94GB HBM3 per GPU	NVL ★★★	NVL ★★★	NVL ★★★				
	H100 PCIe	80GB HBM2e	PCIE ★★★	PCIE ★★★	PCIE ★★★				
	A100	80GB HBM2e	SXM PCIE ★★	SXM PCIE ★★	SXM PCIE ★★				
Graphics/ Compute	L40S	48GB GDDR6	★★	★★★	★	★★★	★★★		★
	L40	48GB GDDR6	★	★★		★★★	★★★		★
	RTX 6000 ADA	48GB GDDR6	★	★★		★★	★★★	★★★	
Small Form Factor Compute /Graphics	L4	24GB GDDR6 72W		★★		★★	★★★	★★★	★★★
	T4	16GB GDDR6 70W		★			★	★	★

*NVL means 2x H100 PCIe paired with NVLink Bridge by default. Regular H100 PCIe and A100 PCIe cards can be paired with NVLink Bridge as option.

How to Configuration??

Hybrid (AI+HPC)

The same platform supports AI (artificial intelligence machine learning training) and HPC (High Performance Computing)

GPU Configuration

The same platform supports various computing power combinations to meet different computing power needs. For example: use H200 for model training and L40S as the inference service.

Crossing

The same platform supports GPU cards of different brands(Nvidia/AMD), reducing the difficult learning curve.

Solution Proposal

AI-Stack :
Computing power scheduling platform

The Issue of Artificial Intelligence

01.Expensive

low utilization of GPUs

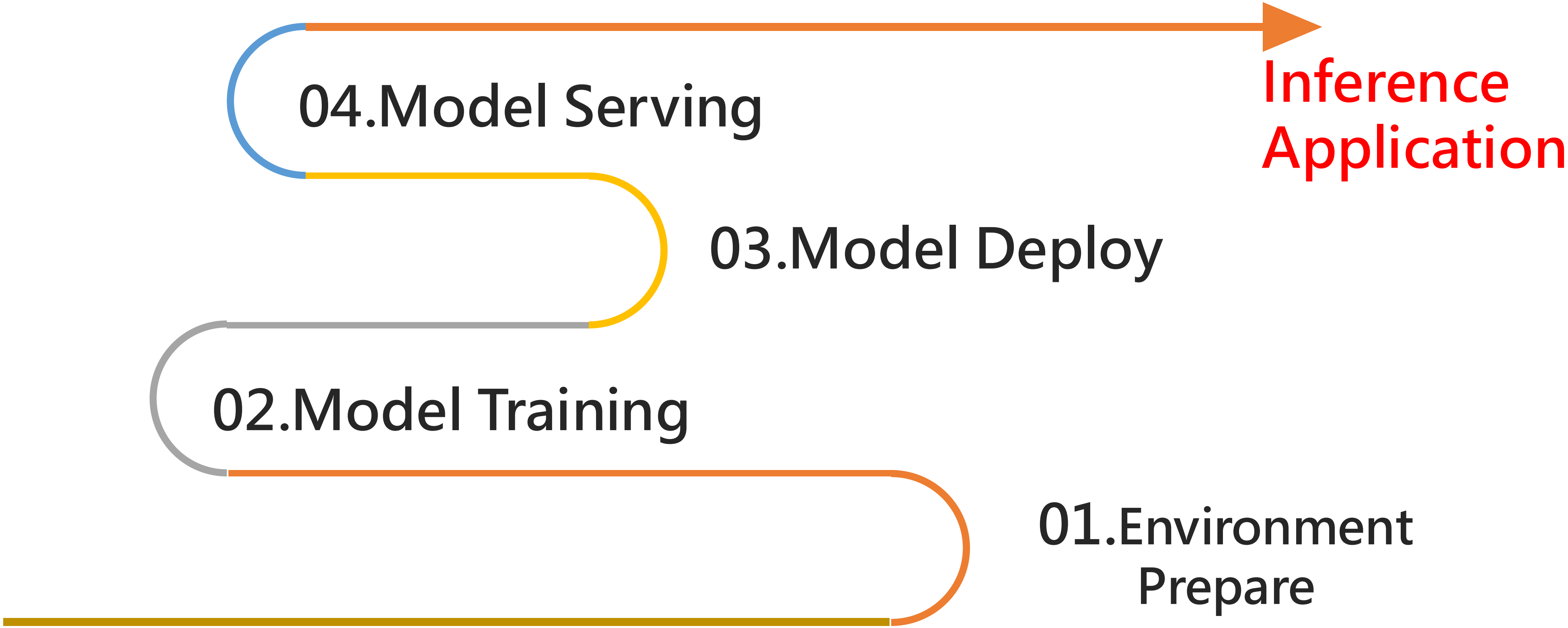
02.Difficult

Difficult to build AI models from 0 to 1 with high accuracy rate

03.Complicated

Complicated in deploying AI to cloud, on-prem and edge

Model Deploy Process



AI-Stack Solution on Artificial Intelligence, AI

Appliance

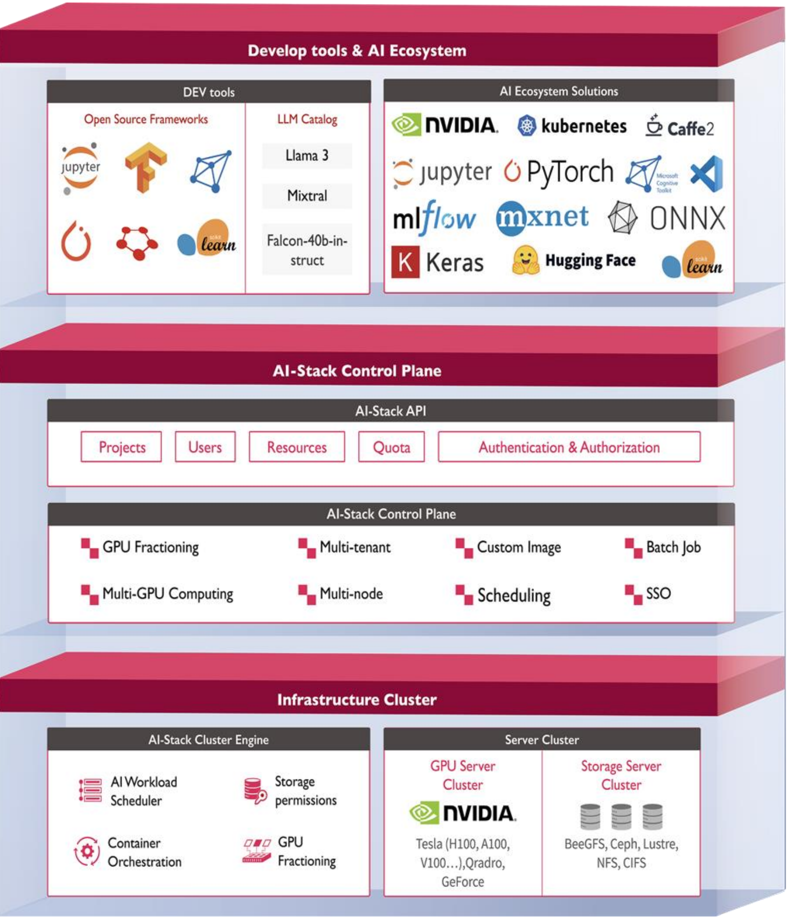
GPU Cluster

GPU Partitioning

LLM/RAG

HPC
Elastic Training

SuperPOD



AI-Stack provides the core infrastructure for AI development

GPU Resource Management
+
MLOps

Help teams manage GPU resources more precisely and efficiently

Seamless connect with commonly used AI development tools

Quota Limit
2500m CPU
Unlimited RAM
4096MiB GPU

Resource isolation, authority management and quota limitations

Automated execution, with options for single or batch tasks

User

Data Scientist/ Developer/ AI Researcher

Data Scientists and AI Researchers
- ✓ Focus on research and model training rather than DevOps
- ✓ Start AI container deployment in just 1 minute

Administrators

IT Administrator

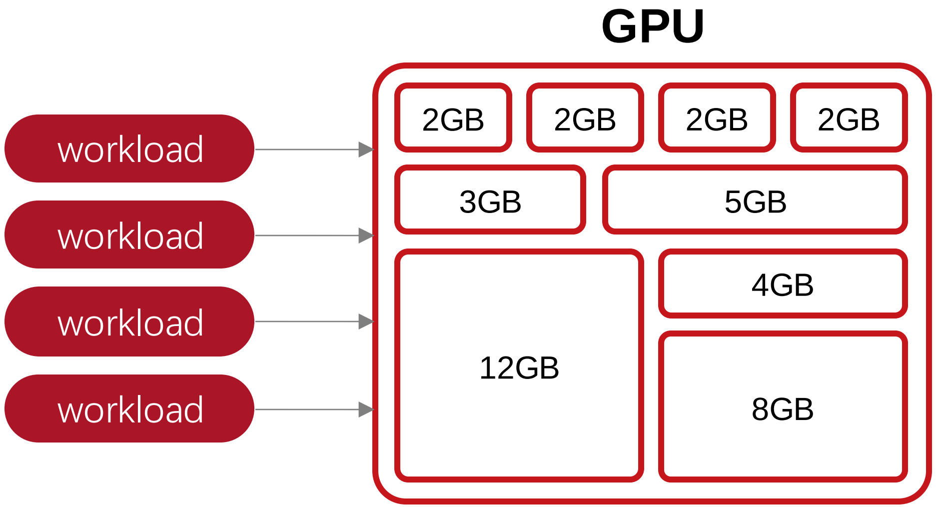
IT Administrators
- ✓ Enterprise-level security protection, efficient resource utilization
- ✓ Easy monitoring and comprehensive resource management

Not all workloads need whole powerful GPUs

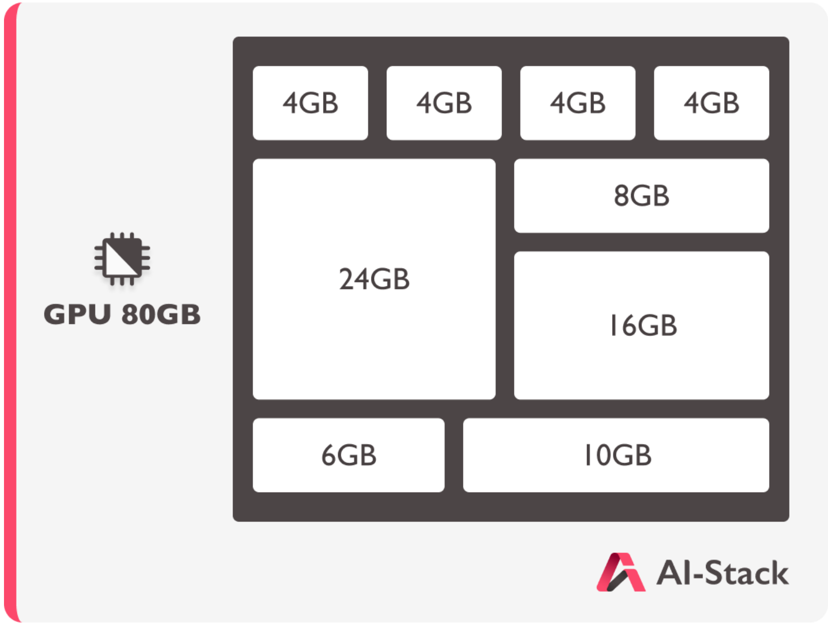
AI-Stack isolates the GPU elasticity into many GPU slices to divide workload requirements.

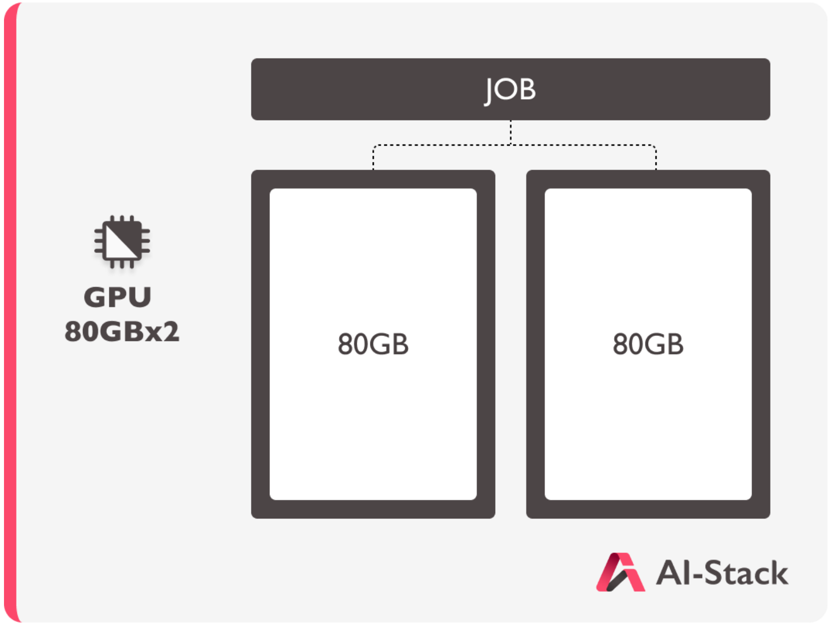
Multiple workloads share a single GPU

- Notebooks
- Inference workloads
- GPU slicing type:
  - AI-Stack GPU Slicing (software isolation)
  - MIG (hardware isolation)



AI-Stack: GPU Partitioning and Aggregation Technology for Maximum Efficiency





Utilization rate ↑

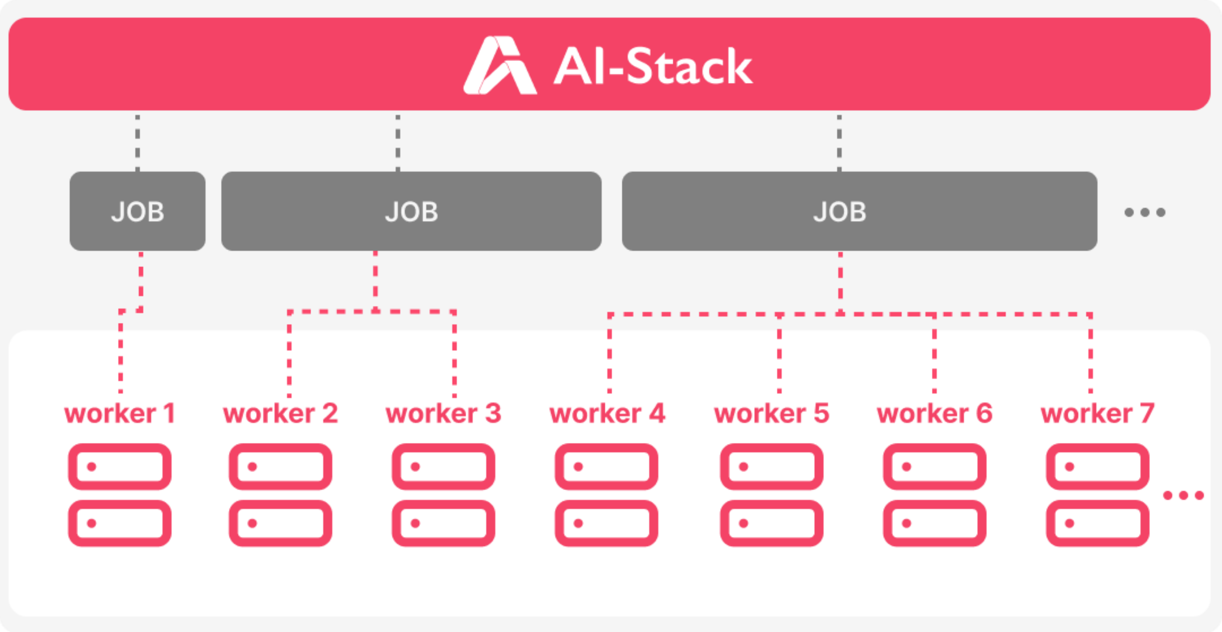
One GPU → Multi GPU Fractions

Computing performance ↑

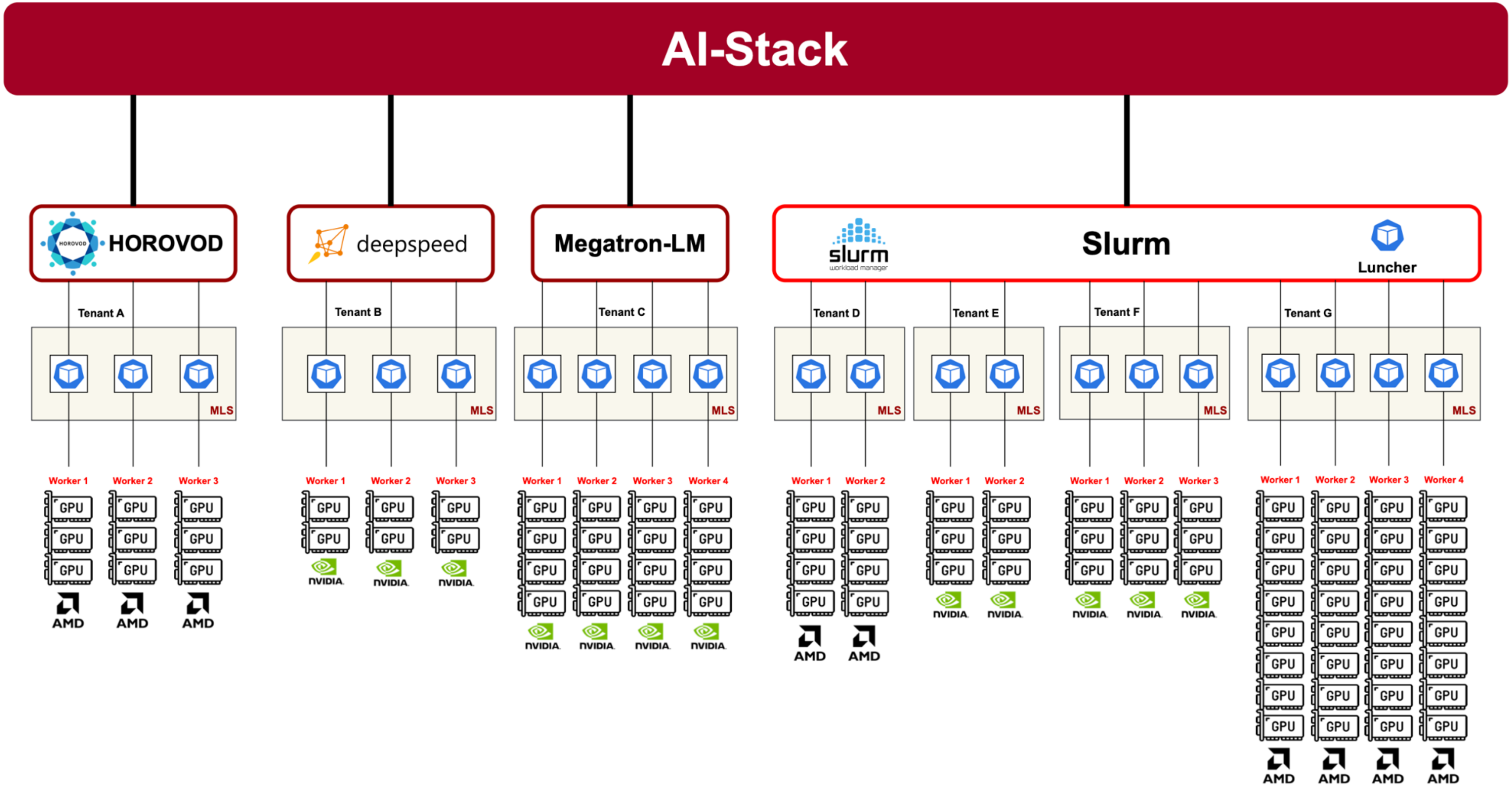
Multi GPUs → One GPU Aggregation

AI-Stack High-performance computing, HPC

The platform can **allocate training tasks across multiple nodes** based on demand and leverage **distributed training** technology to organize multiple containers into training groups. This enables **parallel and distributed processing** of massive datasets, effectively reducing model training time, enhancing computational efficiency, and optimizing resource utilization.



AI-Stack Elastic Training 2.0



GPU Partitioning Example

AI-Stack Slicing	Without AI-Stack
GPUs X 3	GPUs X 12

1
10
10
5
6

V100 32GB

Container
X5

2
20
6
4

V100 32GB

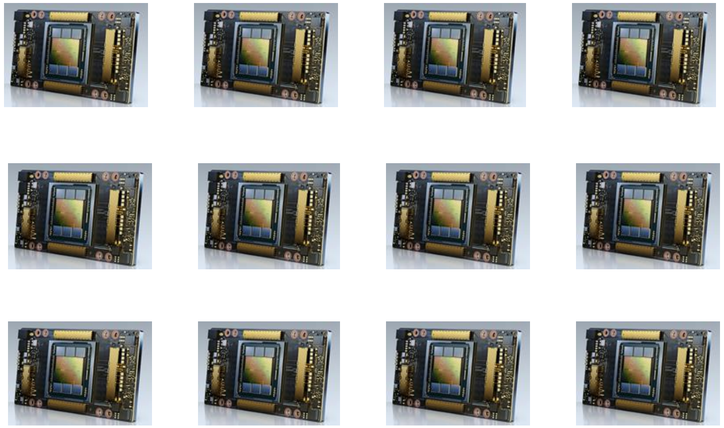
Container
X4

10
16
6

V100 32GB

Container
X3

12 Containers For Training and Inference



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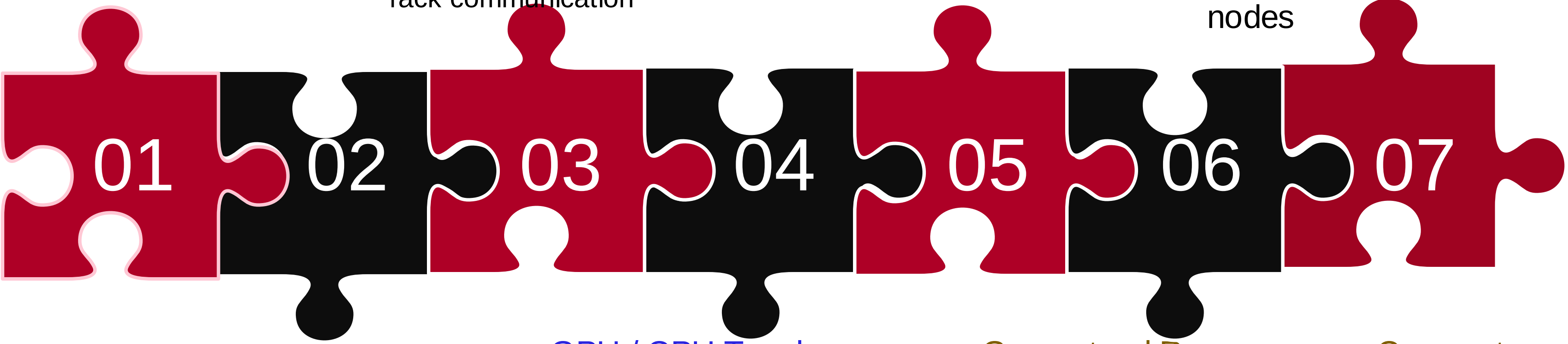
AI-Stack GPU Scheduling Strategy

Gang Scheduling : Schedules dependent tasks simultaneously or delays the entire group

ToR Scheduling : Schedules workloads based on ToR switch topology to reduce inter-rack communication

Load-Aware Scheduling : Assigns tasks to nodes with lower current resource utilization

Balanced Scheduling : Evenly distributes workloads to avoid overloading specific nodes



BinPack Scheduling : Maximizes resource usage by packing workloads densely on fewer nodes

GPU / CPU Topology-Aware Scheduling : Allocates resources based on GPU/CPU interconnects like NVLink or NUMA

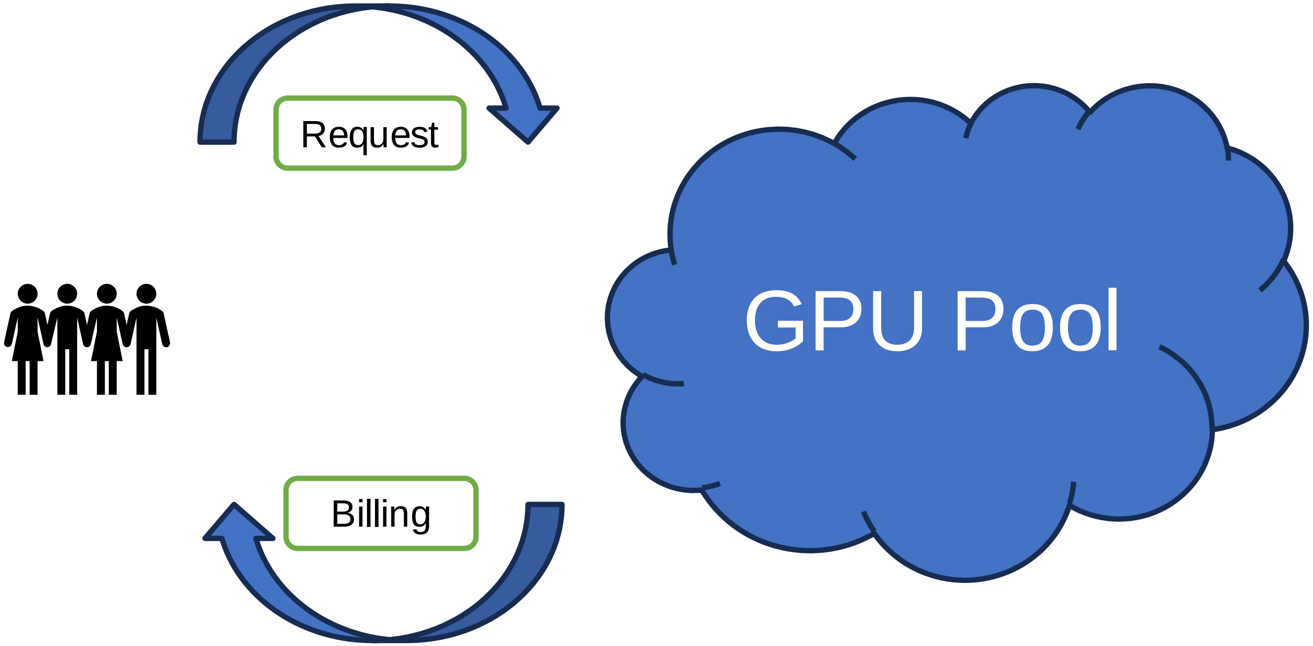
Guaranteed Resource Reservation Scheduling : Reserves resources for critical tasks to ensure availability

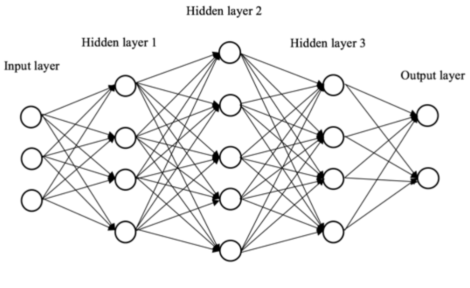
Compact Scheduling : Concentrates workloads on fewer nodes to free idle resources

AI Computing Center

GPU Power Computing Center

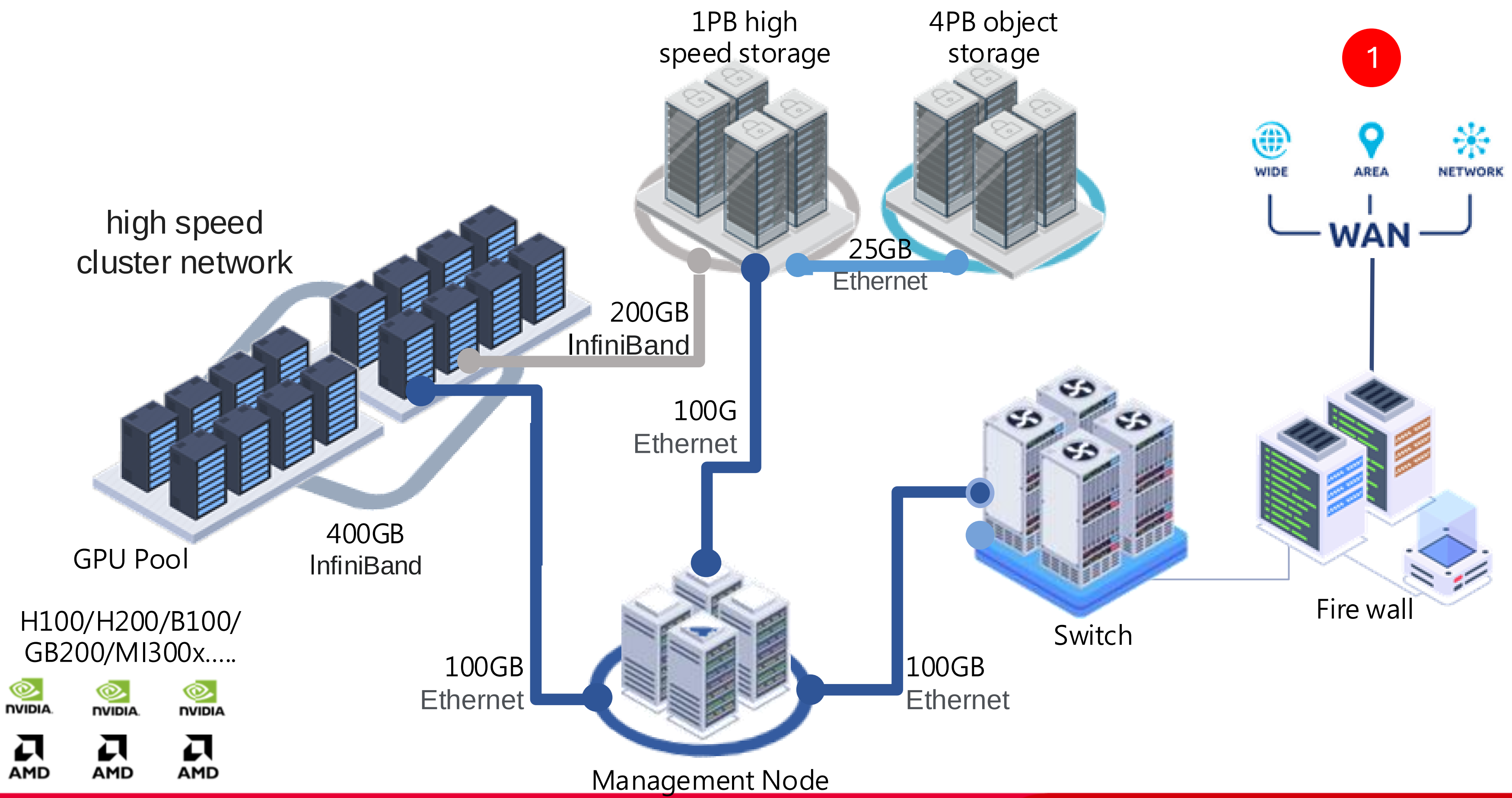
Provide various GPU computing resources to users who need to rent computing power for model training and inference service deployment.





A Deep Learning Model

Power Computing Architecture



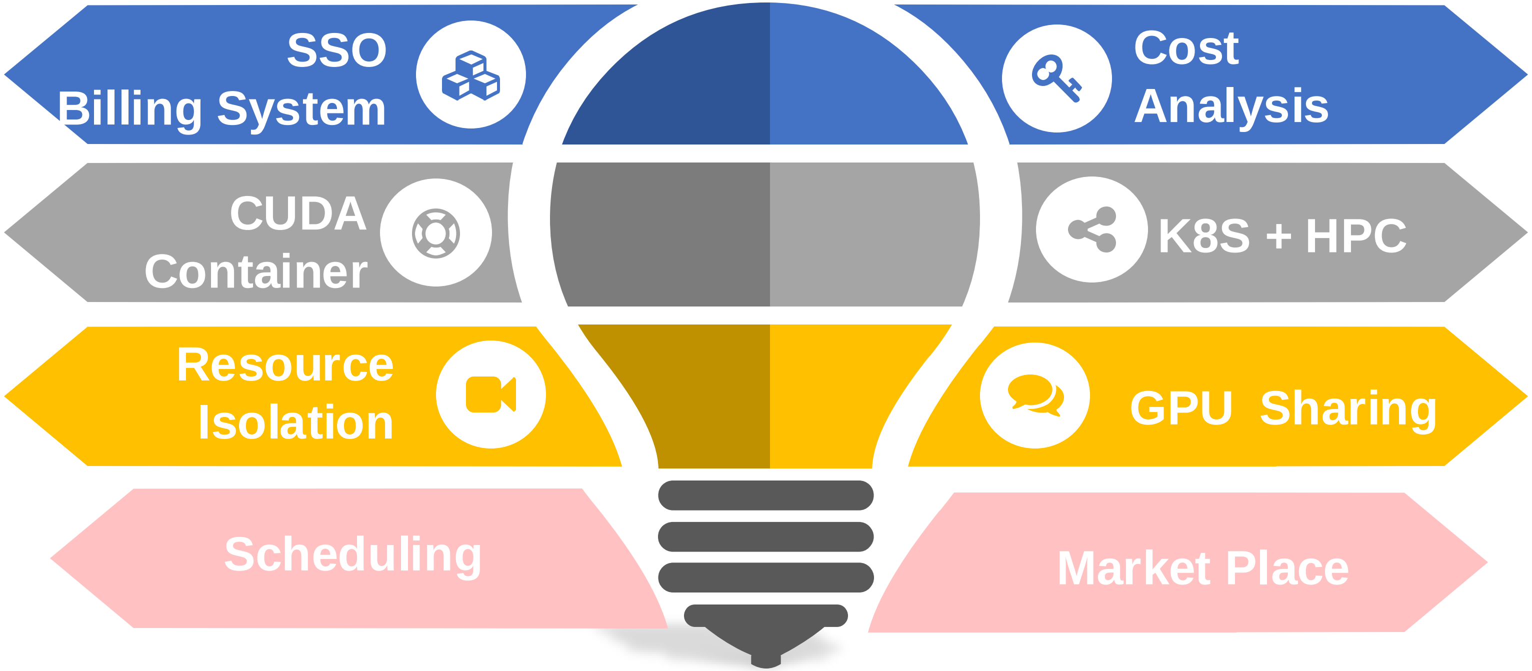
AI-Stack Data Center

Comprehensive Computing Center Solution

AI-Stack Data Center

Developer		Administrator			Operator					
API Catalog	Resource Management (Images, Models)	Billing, Cost Analysis	Access Management	Account Management	Resource Allocation	Dashboard	Health Monitoring, Alert system	Power Consumption and Carbon Emissions Monitoring	Usage Data Analysis	High Availability, Disaster Recovery

Need to know for GPU Computing



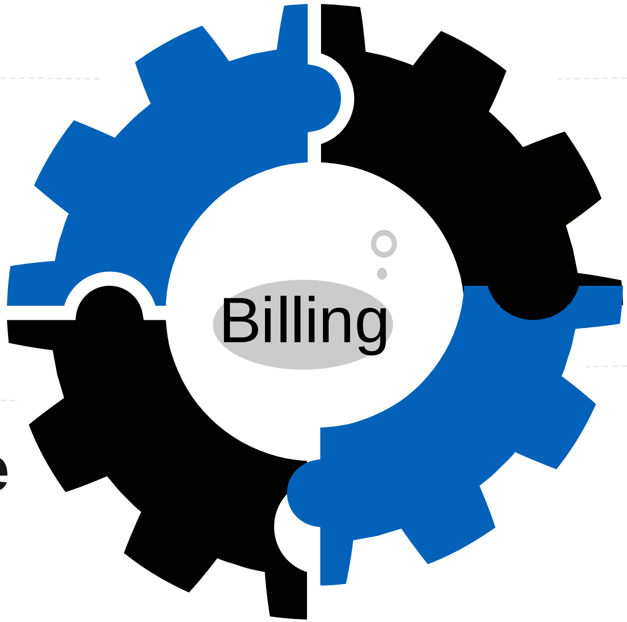
Business Model

Pre-Pay

Advance payment and Debits are made based on usage until the account balance reaches zero

Self-service rental

Computing power rental, including GPU、CPU、Memory and Storage



Monthly Balance

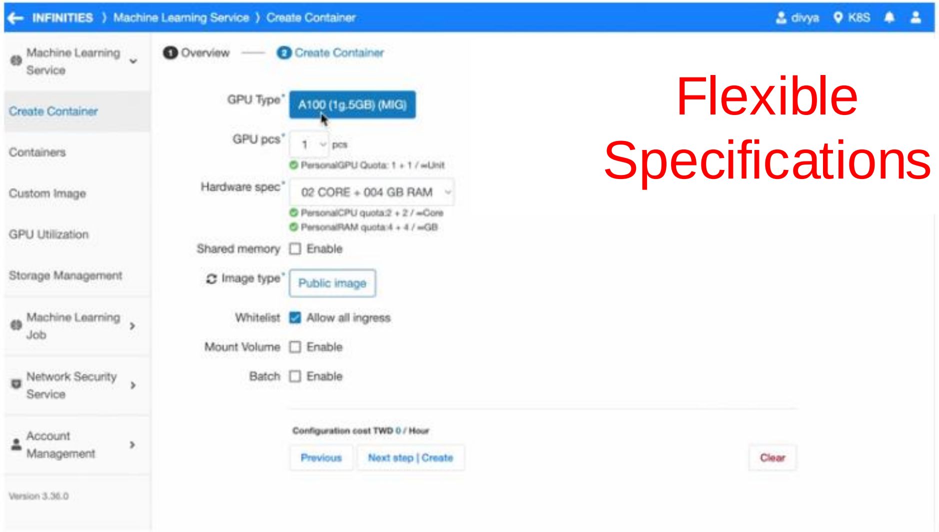
Send bills based on monthly usage

Profit sharing

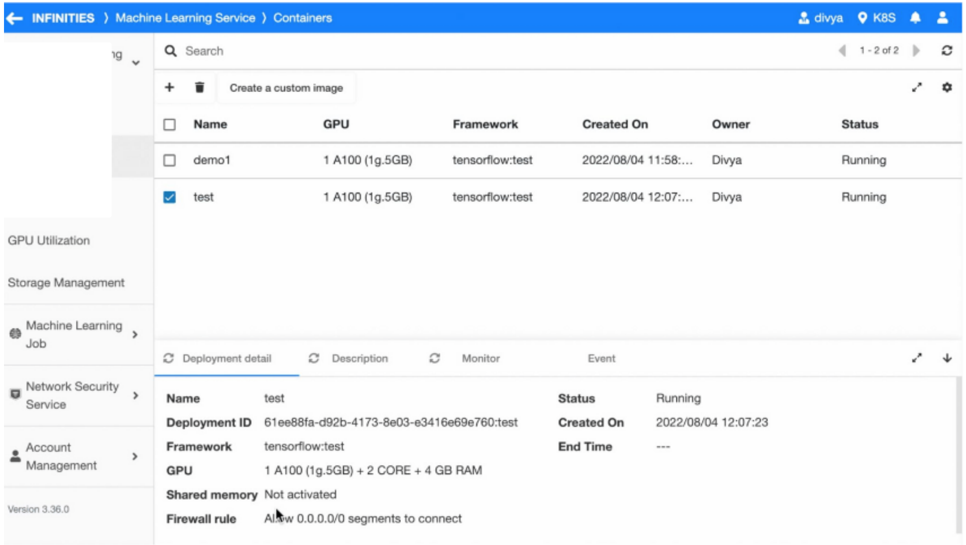
Distribution and agent profit sharing mechanism

AI-Stack Features

User Portal ---Create Container



Flexible Specifications



AI training environment generation in 1 min

User Portal ---Container Service tool

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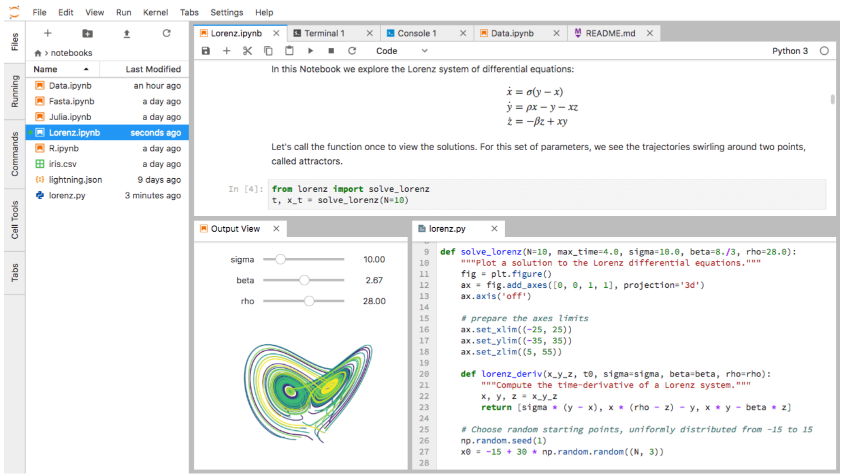
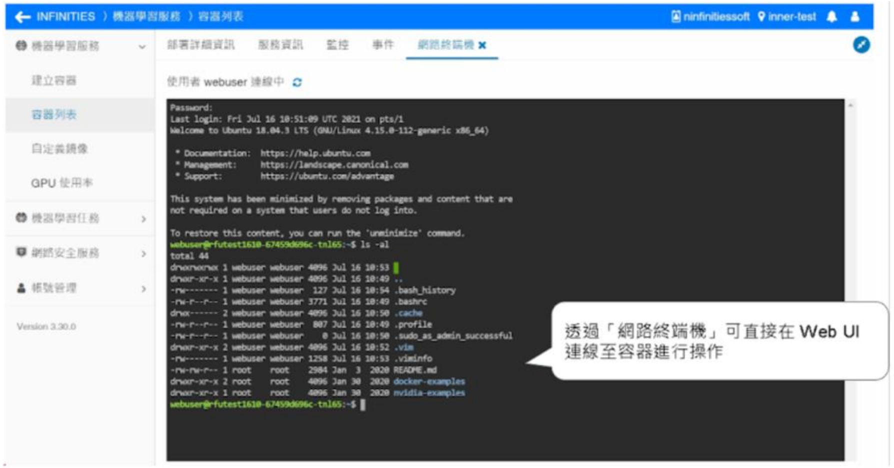




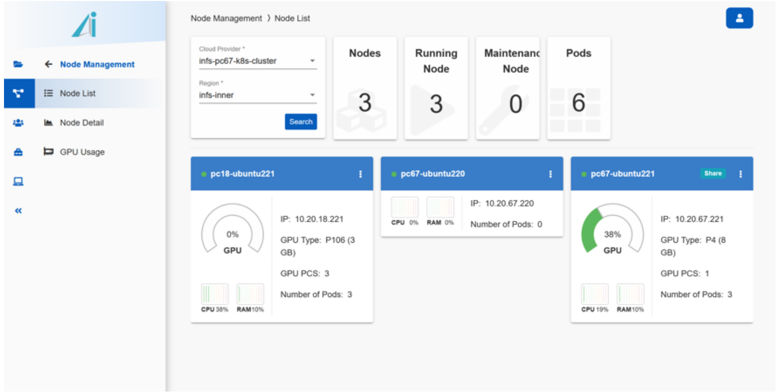
jupyter · jupyterlab · TensorBoard · mlflow · TensorFlow · PyTorch

CODE Visual Studio Code · Keras · MATLAB · mxnet · dmlc XGBoost · H2O.ai

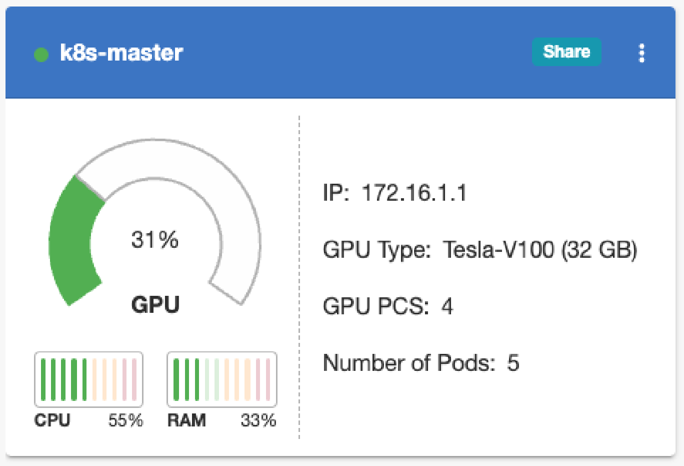
Caffe2 · Caffe · Cognitive Toolkit · scikit learn · ONNX

Administrator Portal

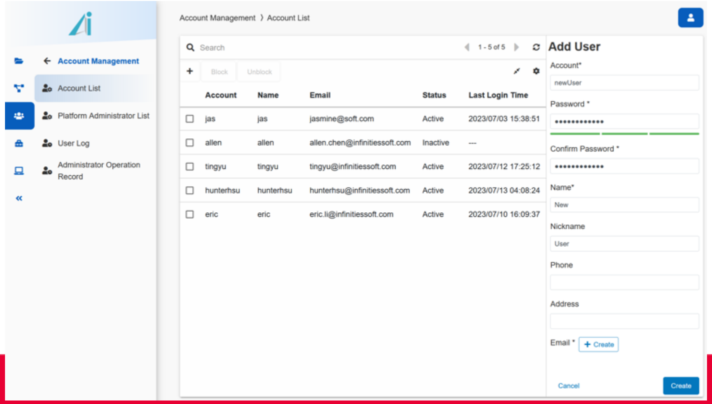
GPU Orchestration



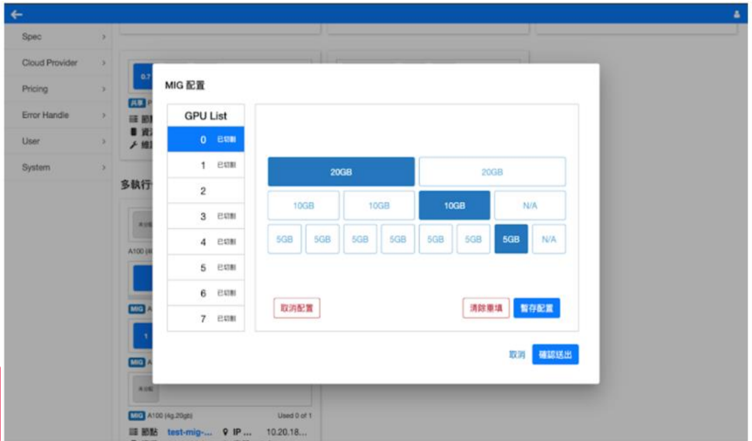
GPU Node Slice setup



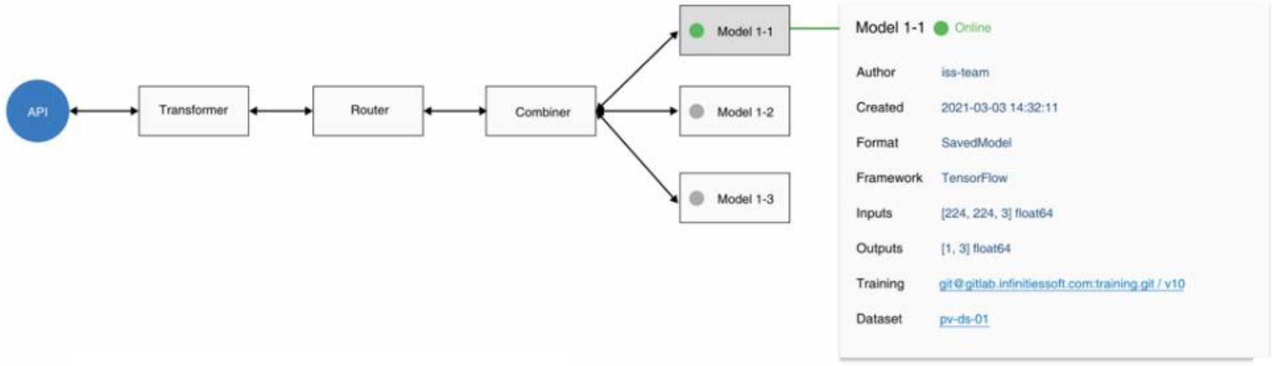
Resource Isolate



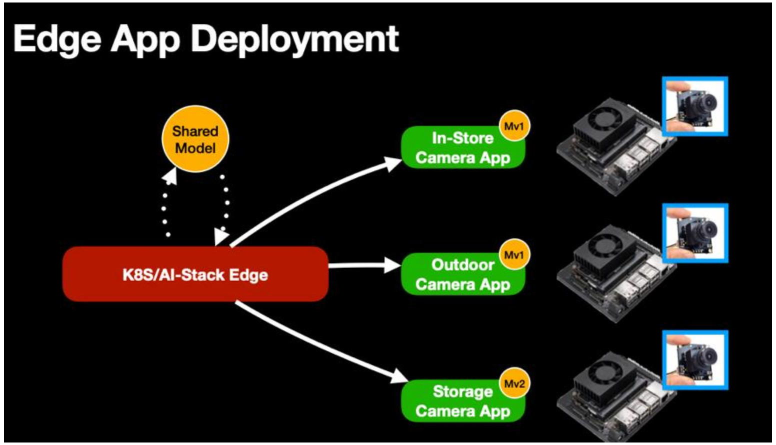
MIG



Deployment

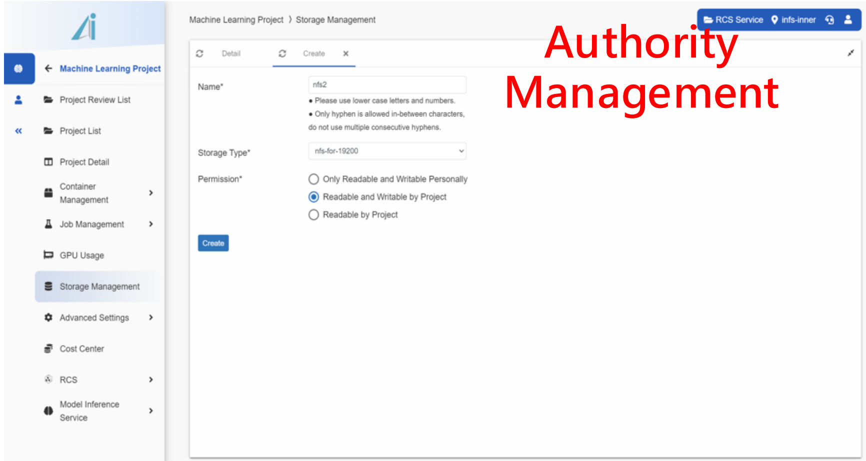


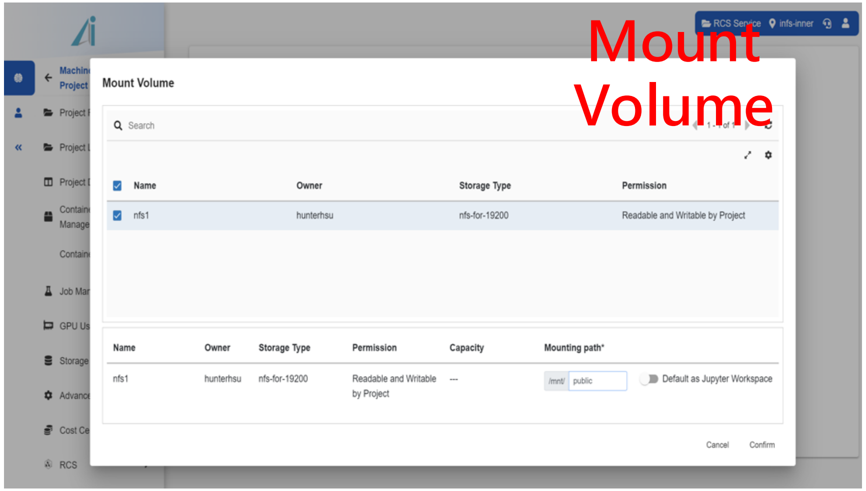
Edge Deployment

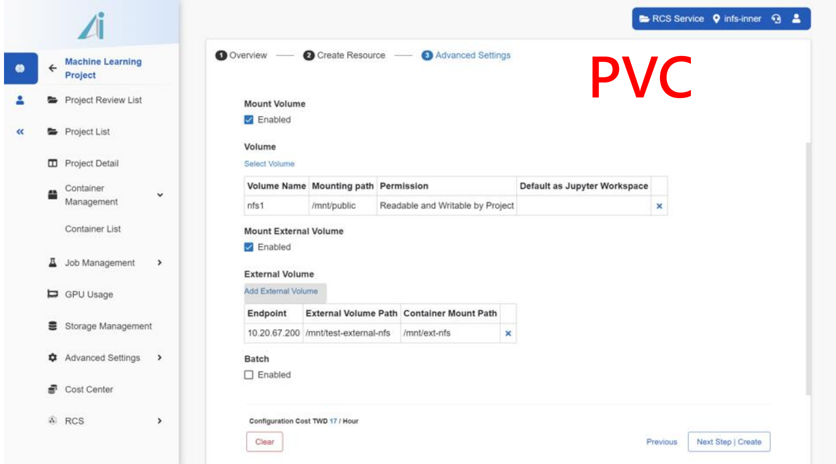


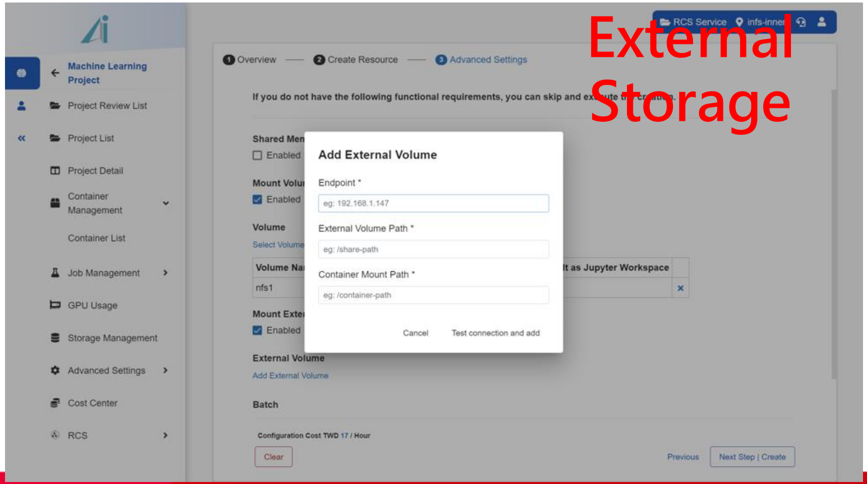
Hyper Link /API

Storage Orchestration









Resource Monitor

Health Monitor

Health Event Monitor

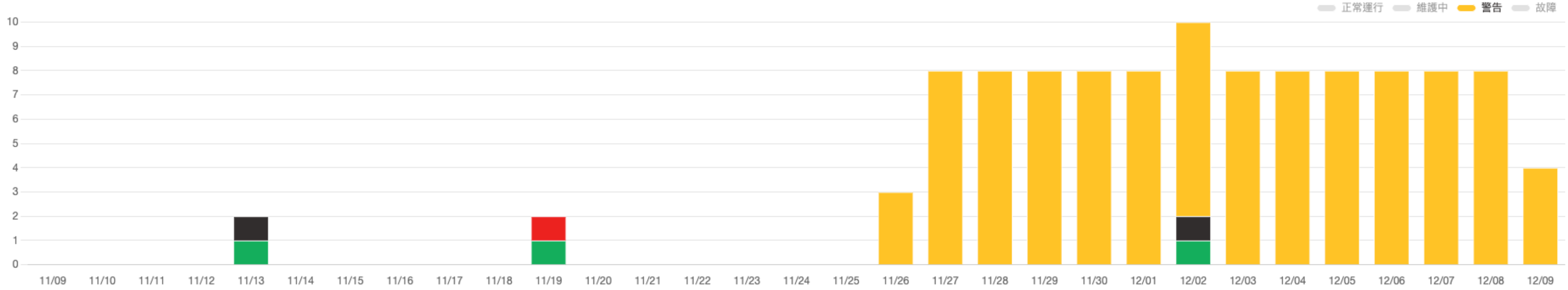
健康狀態監控

叢集： Kubernetes [Kubernetes-infs-inner]

全螢幕

事件發生次數

過去 30 天

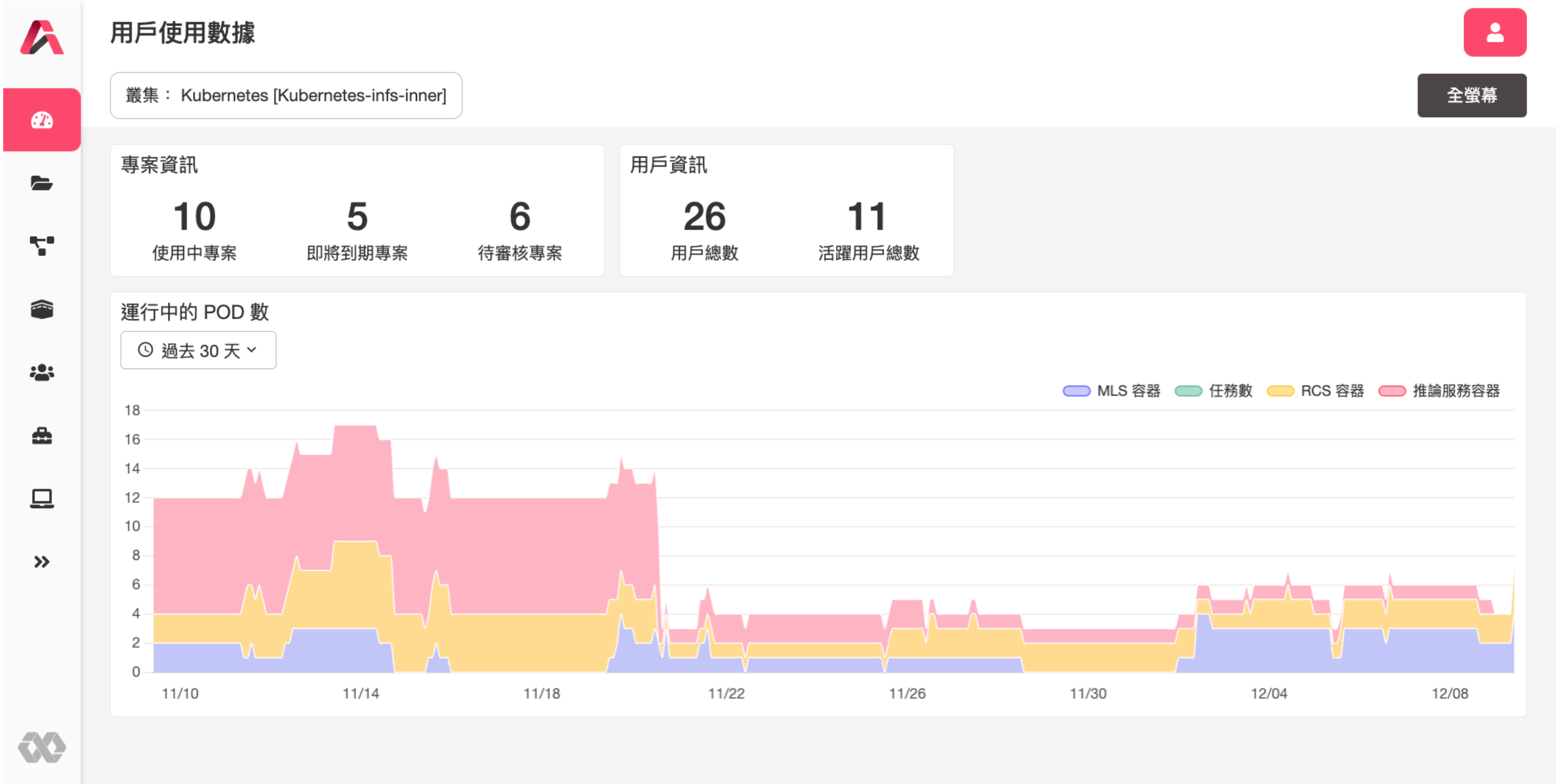


事件紀錄

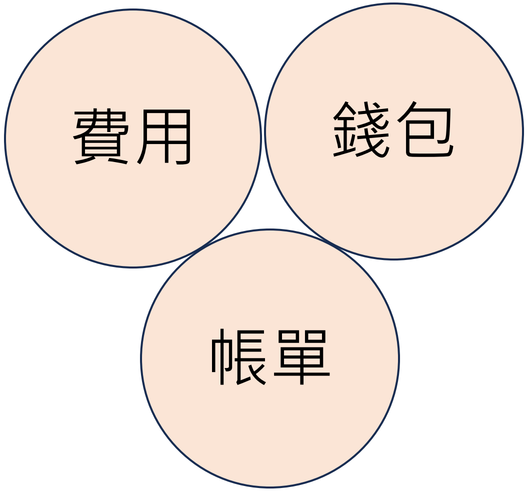
	事件	等級	時間	名稱
1	使用率偏高	警告	2024.11.27 03:51:00	infs-harbor-registry
2	使用率偏高	警告	2024.11.27 00:51:00	infs-harbor-registry
3	使用率偏高	警告	2024.11.26 21:51:00	infs-harbor-registry
4	使用率偏高	警告	2024.11.26 18:51:00	infs-harbor-registry
5	使用率偏高	警告	2024.11.26 15:51:00	infs-harbor-registry
6	恢復正常運行	正常運行	2024.11.19 18:27:00	infs-harbor-registry
7	不健康	故障	2024.11.19 18:25:30	infs-harbor-registry
8	恢復正常運行	正常運行	2024.11.13 15:54:00	pc67-ubuntu221
9	切換為維護模式	維護中	2024.11.13 15:52:56	pc67-ubuntu221

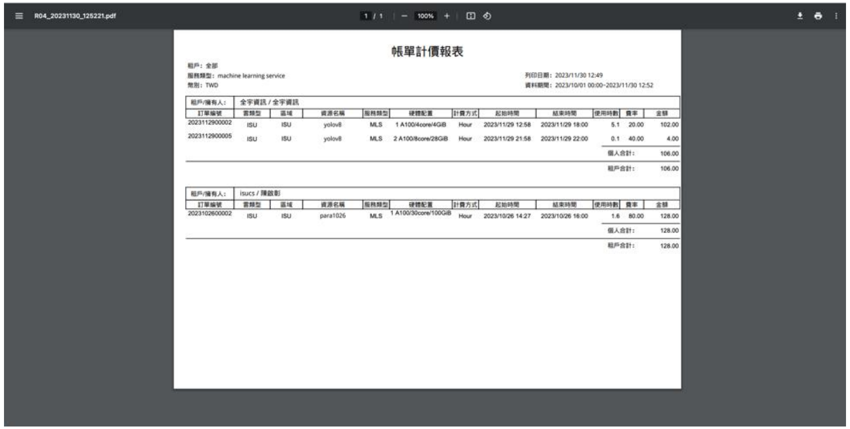
< 1 ... 7 8 9 10 11 >

Utilization Monitor



Expense Analysis





R04_20231130_125221.pdf

帳單計價報表

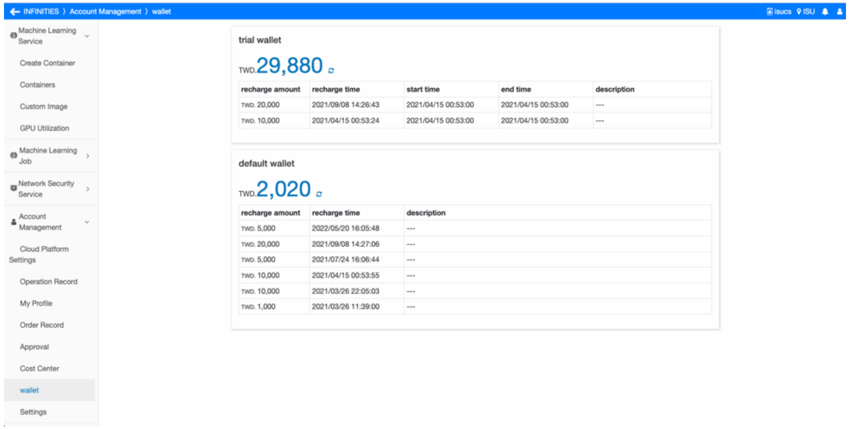
租戶：全部
服務類型：machine learning service
幣別：TWD
列印日期：2023/11/30 12:49
資料期間：2023/10/01 00:00~2023/11/30 12:52

租戶/擁有人：全宇資訊 / 全宇資訊

訂單編號	雲類型	區域	資源名稱	服務類型	硬體配置	計費方式	起始時間	結束時間	使用時數	費率	金額
2023112900002	ISU	ISU	yolov8	MLS	1 A100/4core/4GiB	Hour	2023/11/29 12:58	2023/11/29 18:00	5.1	20.00	102.00
2023112900005	ISU	ISU	yolov8	MLS	2 A100/8core/28GiB	Hour	2023/11/29 21:58	2023/11/29 22:00	0.1	40.00	4.00
										個人合計：	106.00
										租戶合計：	106.00

租戶/擁有人：isucs / 陳啟彰

訂單編號	雲類型	區域	資源名稱	服務類型	硬體配置	計費方式	起始時間	結束時間	使用時數	費率	金額
2023102600002	ISU	ISU	para1026	MLS	1 A100/30core/100GiB	Hour	2023/10/26 14:27	2023/10/26 16:00	1.6	80.00	128.00
										個人合計：	128.00
										租戶合計：	128.00



INFINITIX

AI-Stack Software Stack Roadmap

AI Enteriprise Software Stack

AI Marketplace (AI-Stack)

AI and Data Science Tools/ Framework

- Industry Solutions
- Helm Charts
- Model Scripts
- Pre-trained Models
- Containers
- NGC import

Ready-to-use, fully supported software that **speeds developer success**

AI-Stack Platform

DEV AI Lifecycle integration	CONTROL PLANE Resource Pooling Policy Engine	CLUSTER ENGINE Workload Orchestration GPU Partitioning
MLOps	Auto Aware Scheduling	Kubernetes
IDE, Tracking, Pipeline, LLM, Training / Serving	Gang, Binpack, ToR, Topology, Load, Auto	Slrum, MPI, Elastic Training, GPU Partitioning & Aggregation

Get more models from **prototype to production**

Ensure hassle-free **execution of every developer's jobs**

Base Command (AI-Stack)

Cluster Management (AI-Stack)

- ENERGY
- PROVISIONING
- MONITORING
- CLUSTERING *
- MANAGEMENT

Effortlessly **scale and manage** one node to thousands

Network / Storage Acceleration Libraries & Management

- NETWORK IO
- STORAGE IO
- IN-NETWORK COMPUTE
- IO MANAGEMENT

Accelerate end to end **infrastructure performance**

Operating System

OS Extensions for Linux Distributions

Maximize system uptime security and reliability

A設備

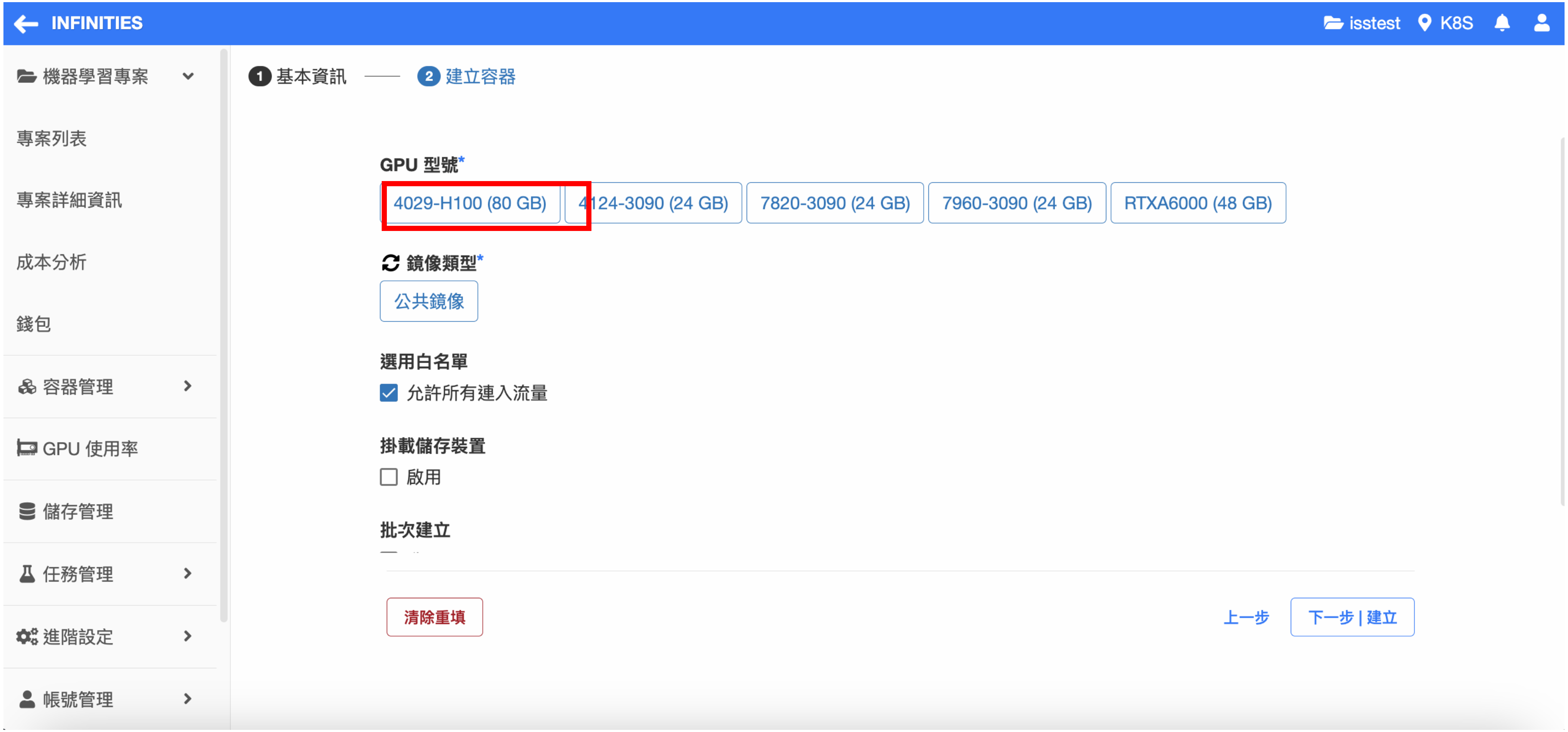
B設備

C設備

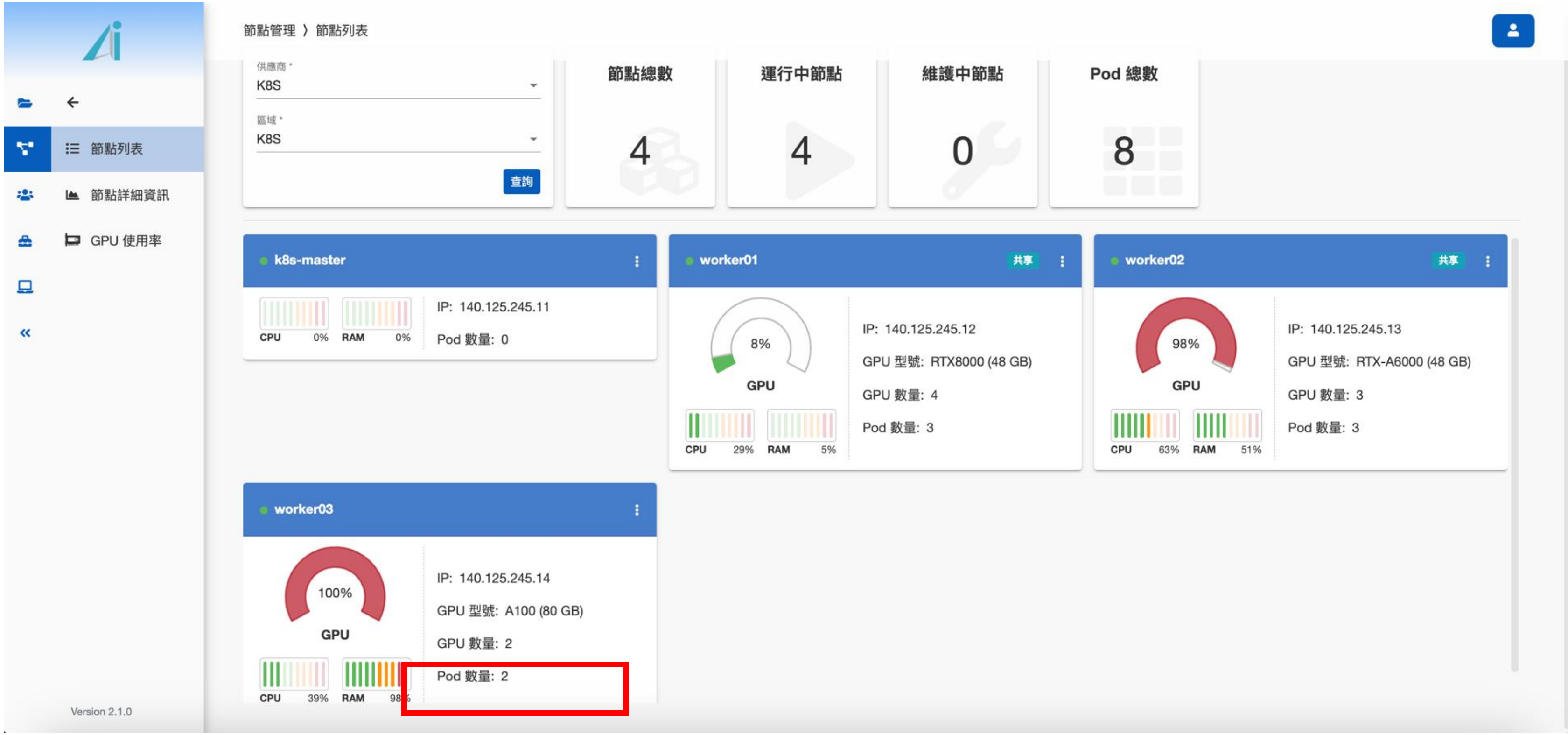
D設備

GPU Pooling

Nvidia GPU Pool (H100+RTX3090+RTXA6000)



Nvidia GPU Pool (A100+RTX8000+RTX A6000)

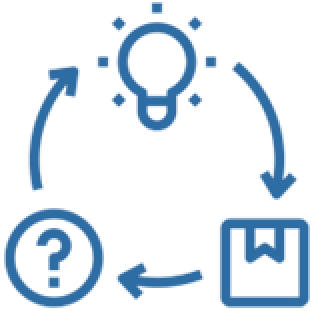


Nvidia GPU Pool(DGX-1 A100)

DGX H100*4 + AMD MI300X

Why AI-Stack

Support for the entire AI lifecycle

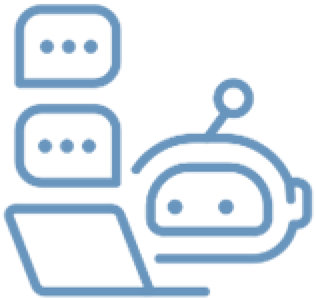


Model Development

Dev & debug in IDE tools like Jupyter notebooks, VSCode, PyCharm etc

Fine-tuning & Training

Run long model tuning or training workloads as batch jobs



Prompt Engineering

Experiment with language and GenAI models through prompt engineering

Serving in Production

Deploy models in production to serve business applications

55

Total Cost of Ownership

	AI-Stack	Without AI-Stack	Description
Environment Generation	Easy (60s)	Complex (28,800s)	Without AI-Stack, Advanced users need one day to prepare the model training environment.
Cooperation	Easy	X	AI-Stack supports a large number of users to log in to the system at the same time for model training and inference serving
Model as Service	Easy	Complex	Without AI-Stack, user need to packing and deploy model as service manually
Maintenance	Easy	Complex	Without AI-Stack, user need Kubernetes skill to maintenance and management GPU cluster
GPU Utilization	High (90%)	Low (30%)	AI-Stack provide GPU slices and job scheduling to upper GPU utilization
TCO	**Low**	High	

AI-Stack enhances GPU efficiency when helping enterprises implement AI

90% ↑

GPU Utilization

10x ↑

Workload execution

1min ↓

Development Environment Setup

10x ↑

Enhanced ROI

GPU Utilization

GPU Partitioning Increases Utilization 30% → 90%

Workload execution

Multiple users and tasks increase efficiency by 10x

DevEnv Setup

Reduces setup time from 2 weeks → 1 min

Enhanced ROI

Boosts return on investment by 10x

Infinitix and AI-Stack is your BEST choice.

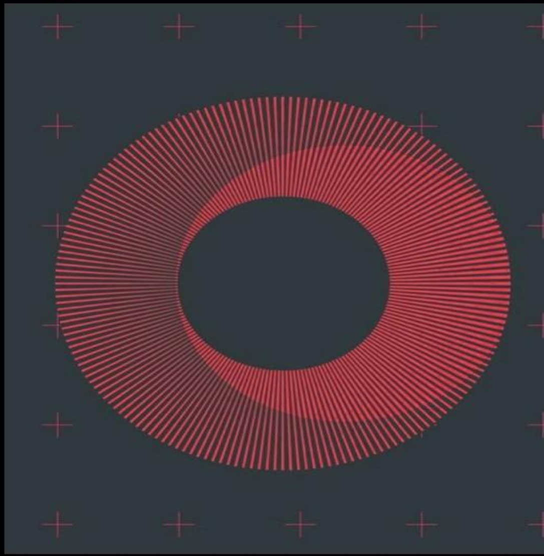
- Open platform, NO Vendor LOCK-IN
- High integrative
- Proved experiencing
- Reliable technical team
- Most cost-effective

INFINITIX

領先
Lead

創新
Innovative

永續
Sustainability



開放
Open